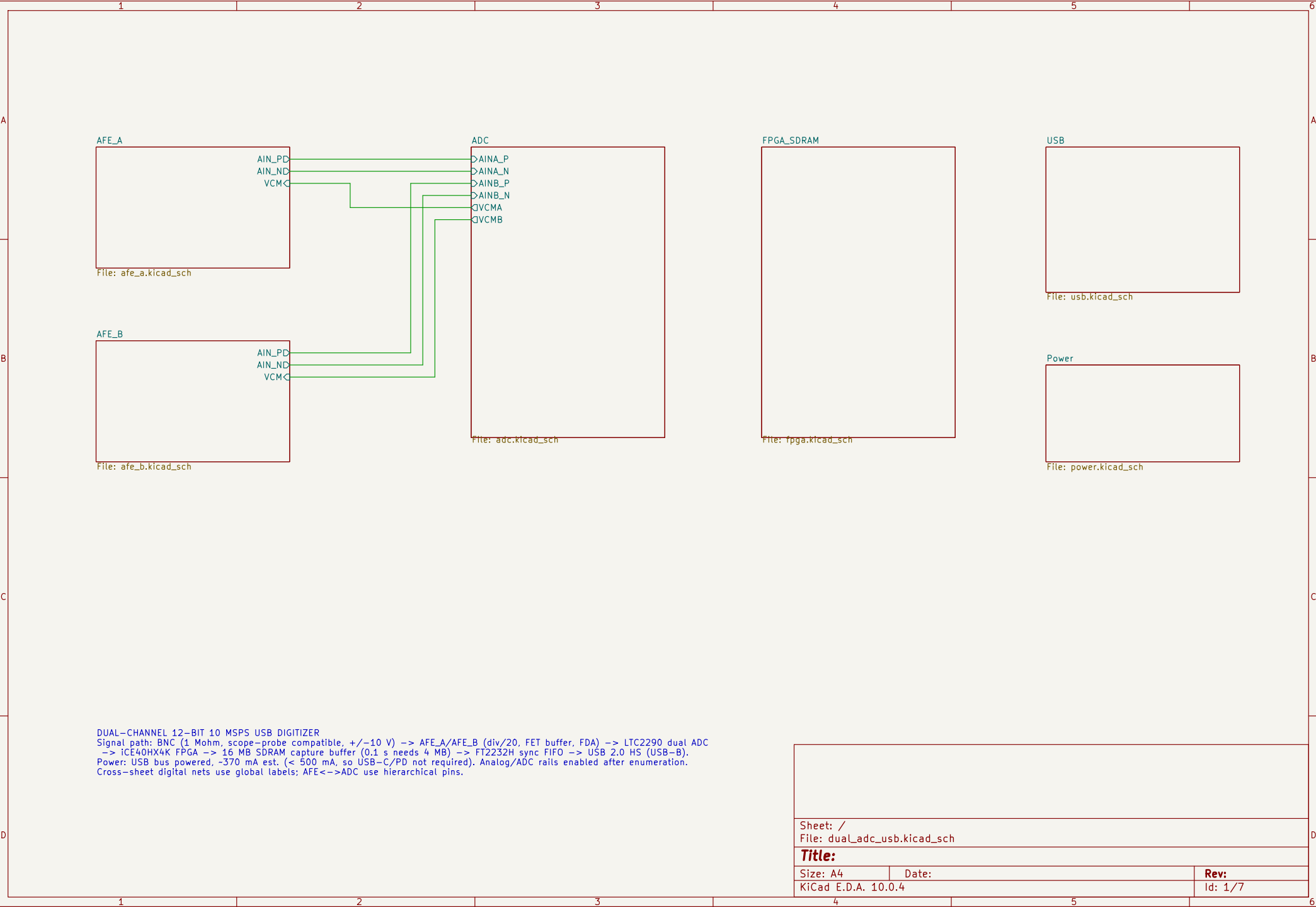
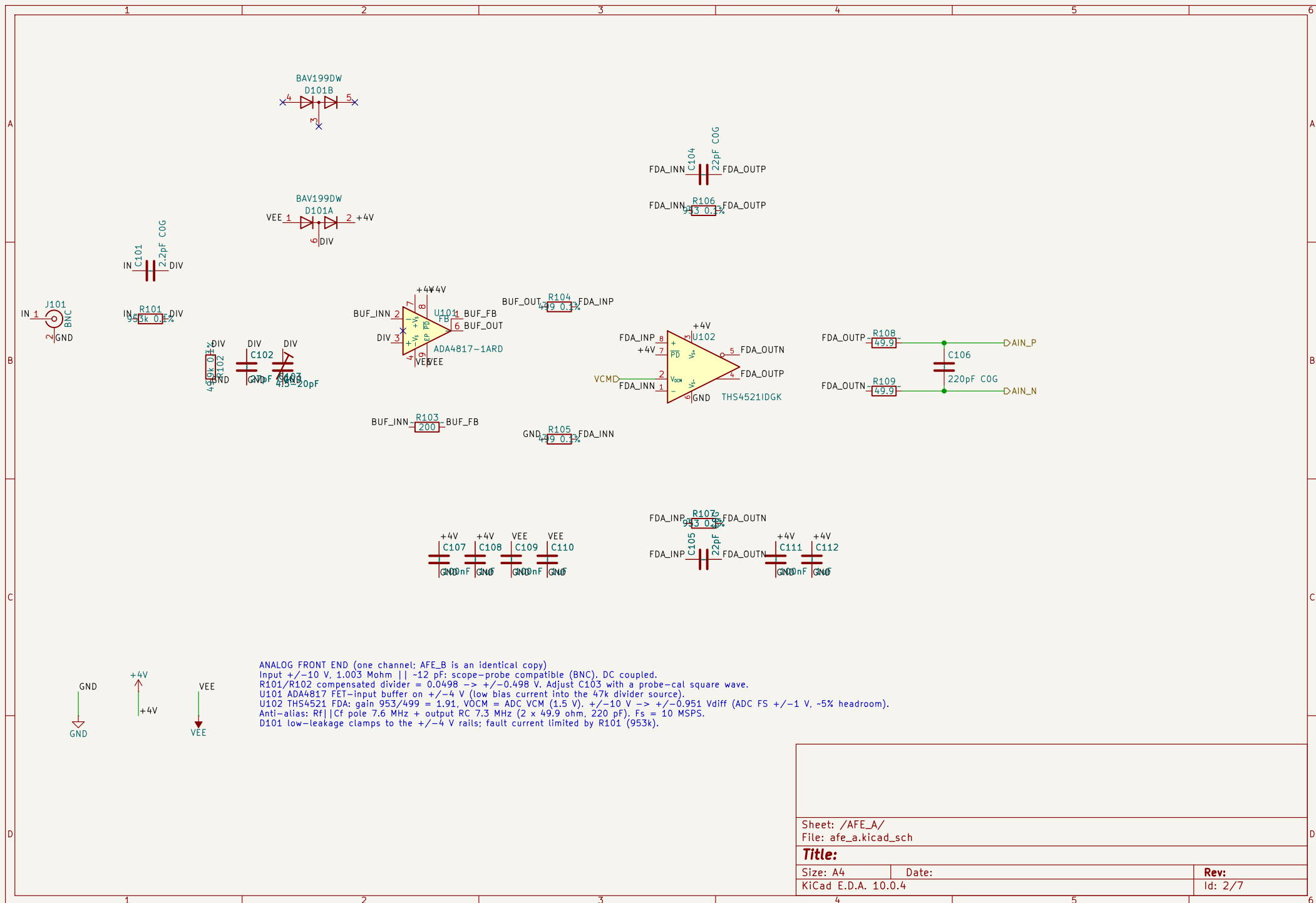






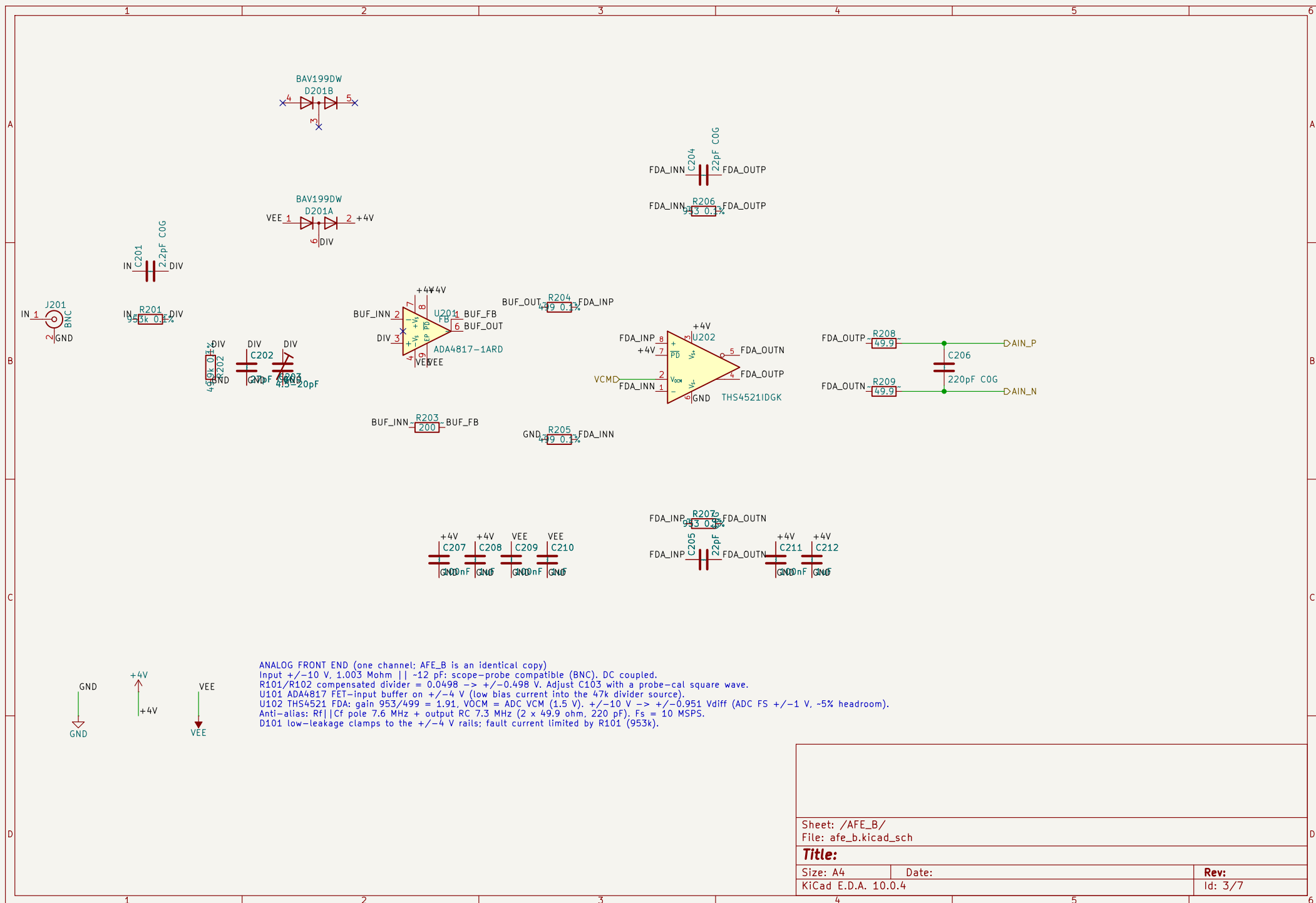
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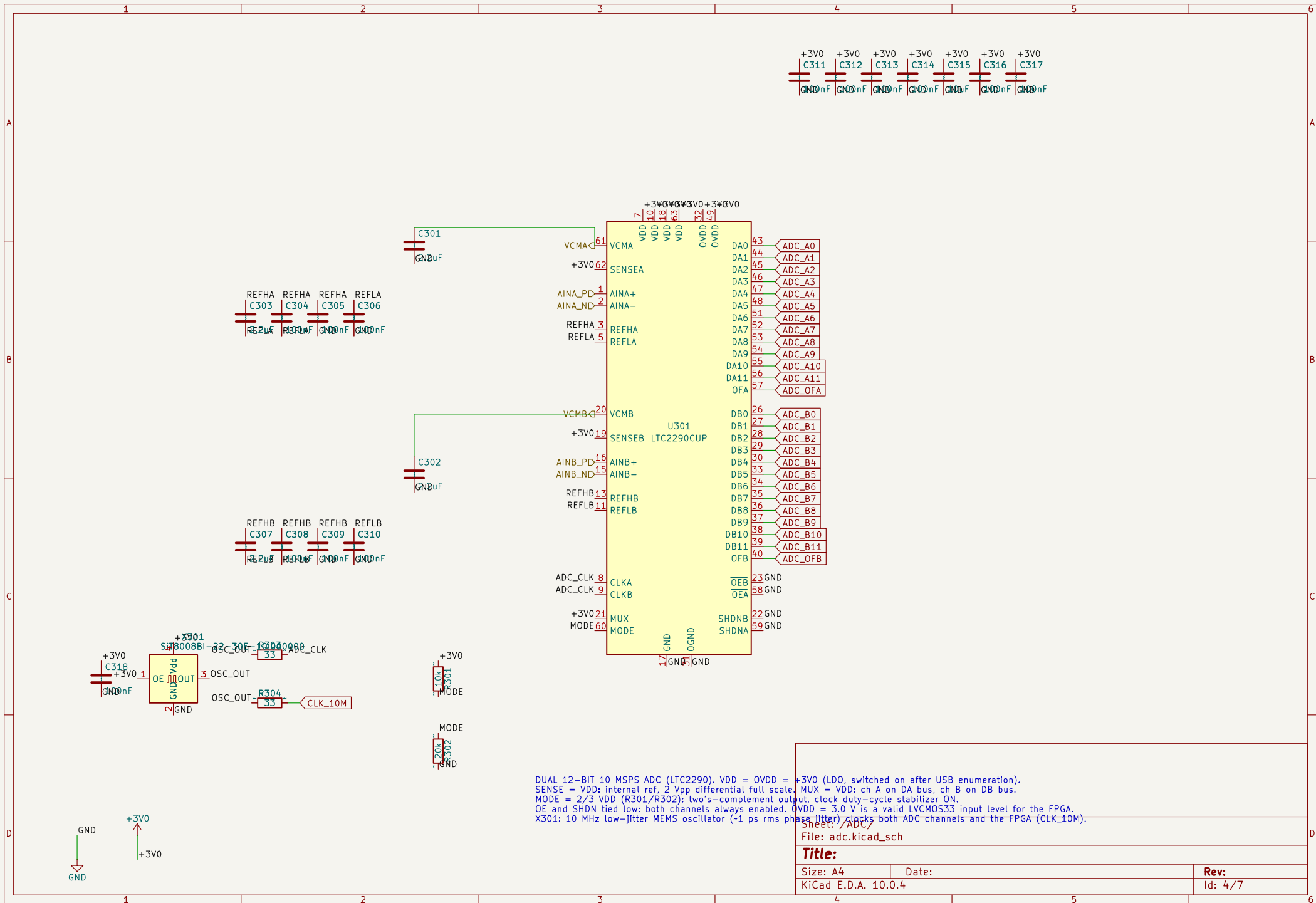
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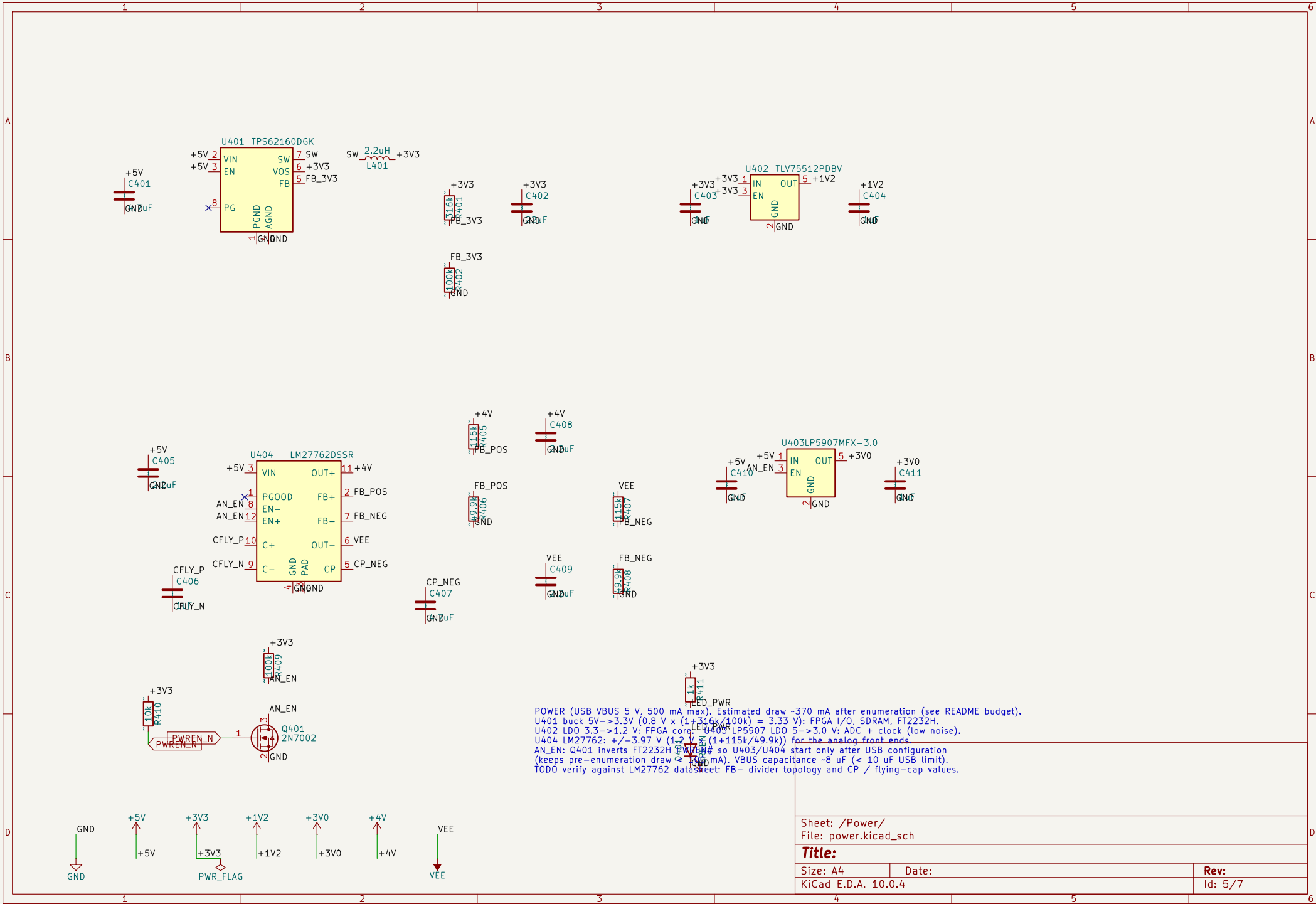
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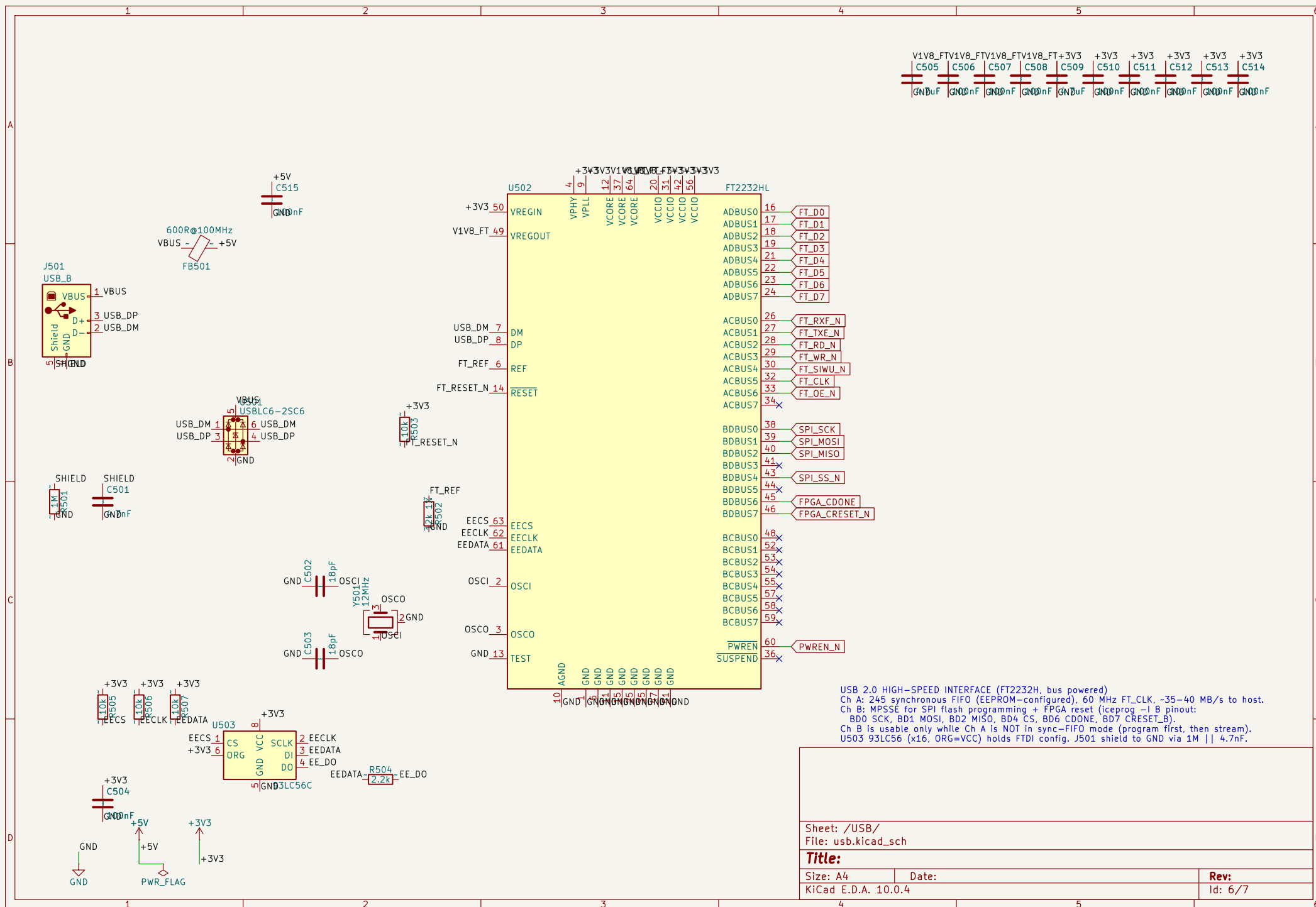
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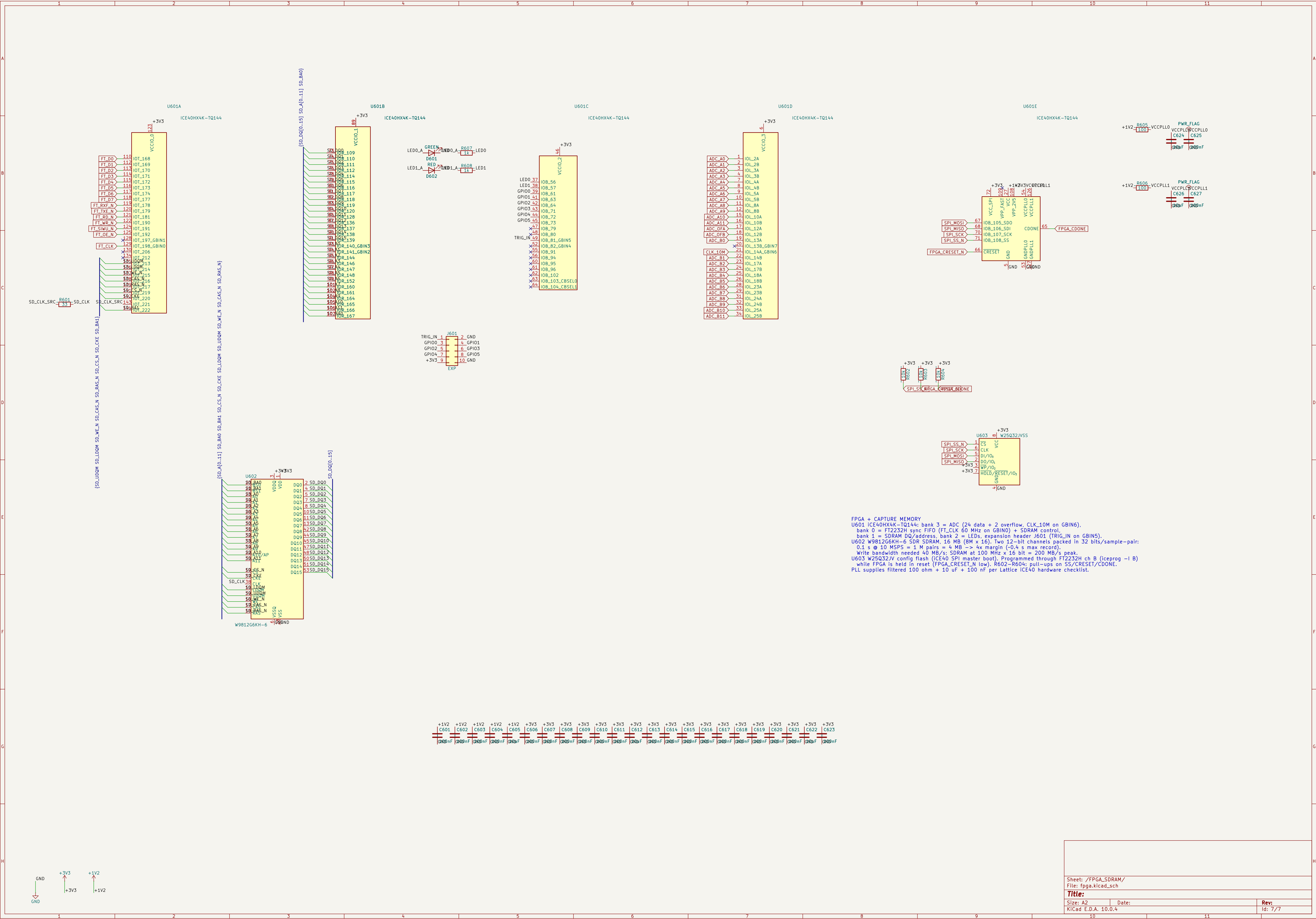
Rev:
Id: 2/7











FPGA + CAPTURE MEMORY
U601 ICE40HX4K-TQ144: bank 3 = ADC (24 data + 2 overflow, CLK_10M on GBIN6), bank 0 = FT232H sync FIFO (FT_CLK 60 MHz on GBIN0) + SDRAM control, bank 1 = SDRAM DQ/address, bank 2 = LEDs, expansion header J601 (TRIG_IN on GBIN5).
U602 W9812G6KH-6 SDR SDRAM, 16 MB (8M x 16), two 12-bit channels packed in 32 bits/sample-pair: 0.1 s @ 10 MSPS = 1 M pairs = 4 MB -> 4x margin (-0.4 s max record).
Write bandwidth needed 40 MB/s; SDRAM at 100 MHz x 16 bit = 200 MB/s peak.
U603 W25Q32JV config flash (ICE40 SPI master boot). Programmed through FT232H ch B (iceprog -l B) while FPGA is held in reset (FPGA_CRESET_N low). R602-R604: pull-ups on SS/RESET/CDONE.
PLL supplies filtered 100 ohm + 10 uF + 100 nF per Lattice ICE40 hardware checklist.